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Visibility obstruction detection for a vehicle

Abstract

A vehicle includes a window separating a compartment of the vehicle from a region external to the vehicle. The window includes an outer surface. An imaging device is in the compartment and configured to capture an image of the region external to the vehicle through the window. Control circuitry is in communication with the imaging device and configured to receive the image from the imaging device, detect an obstruction in the image data corresponding to a substance on the window, and generate an output in response to the detection of the obstruction, wherein the imaging device is disposed within 50 millimeters (mm) from the window.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to image processing in a vehicle environment and, more particularly, to visibility obstruction detection for a vehicle.

BACKGROUND OF THE DISCLOSURE

(2) Conventional image processing techniques for detecting moisture conditions on or around a vehicle may be limited.

SUMMARY OF THE DISCLOSURE

(3) According to a first aspect of the present disclosure, a vehicle includes a window separating a

compartment of the vehicle from a region external to the vehicle. The vehicle includes an imaging device in the compartment and configured to capture an image of the region external to the vehicle through the window. The vehicle includes control circuitry in communication with the imaging device. The control circuitry is configured to receive the image from the imaging device, process the image in a fully-connected data description neural network (FCDDNN) trained to determine a moisture level, detect an obstruction in the image, correlate the obstruction with a substance on the window, classify the substance as water or debris based on the moisture level and generate an output in response to detection of the obstruction.

(4) Embodiments of the first aspect of the present disclosure can include any one or a combination of the following features: a wiper selectively moveable along the window and a window clearing system including a controller in communication with the control circuitry and configured to operate the wiper in response to the output; the window clearing system includes a pump for pressurizing cleaning fluid to apply to the window, wherein the controller is configured to selectively operate the pump in response to the output; the output generated by the control circuitry is based on the classification of the obstruction as water or debris; the controller is configured to operate the wiper in response to classification of the substance as water and operate the pump in response to classification of the substance as debris; the imaging device is disposed within 50 millimeters (mm) from the window; the control circuitry includes a training module that trains the FCDDNN based on manual operation feedback; the manual operation feedback includes manual operation of at least one switch controlling at least one of the wiper and the pump; the training module is configured to train the FCDDNN to differentiate between light sources in the region external and the substance on the window; the control circuitry is configured to classify a level of optical distortion with the moisture level of the substance; the window is a windshield of the vehicle extending at an oblique angle relative to a lens of the imaging device the imaging device is disposed within 15 millimeters of the window; the control circuitry is configured to detect another vehicle ahead of the vehicle; and classify the substance as tire splash from the other vehicle; and the control circuitry, wherein the automatic speed control system is configured to control the speed of the vehicle based on classification of the substance as tire splash from the other vehicle.

(5) According to a second aspect of the present disclosure, a vehicle includes a windshield separating a compartment of the vehicle from a region external to the vehicle. The vehicle includes an imaging device in the compartment and is configured to capture an image of the region external to the vehicle through the windshield and control circuitry in communication with the imaging device. The control circuitry is configured to receive the image from the imaging device, detect an obstruction in the image corresponding to a substance on the windshield, and generate an output in response to the detection of the obstruction. The imaging device is disposed within 50 millimeters (mm) from the windshield.

(6) Embodiments of the second aspect of the present disclosure can include any one or a combination of the following features: a wiper selectively moveable along the windshield and window clearing system including a controller in communication with the control circuitry and configured to operate the wiper in response to the output; the imaging device is disposed within 15 mm of the windshield; and the control circuitry is configured to classify a level of optical distortion with the moisture level of the substance.

(7) According to a third aspect of the present disclosure, a vehicle includes a windshield separating a compartment of the vehicle from a region external to the vehicle. The vehicle includes a wiper selectively moveable along the windshield. The vehicle includes a window clearing system including a controller configured to operate the wiper. The vehicle includes an imaging device in the compartment and configured to capture an image of the region external to the vehicle through the windshield. The imaging device is disposed within 50 millimeters (mm) from the windshield and includes a lens. The windshield extends at an oblique angle relative to the lens. The vehicle includes control circuitry in communication with the imaging device and the window clearing

system. The control circuitry configured to receive the image from the imaging device, process the image in a fully-connected data description neural network (FCDDNN) trained to determine a moisture level, detect an obstruction in the image, correlate the obstruction with a substance on the windshield, classify the substance as water or debris based on the moisture level, and generate an output in response to classification of the substance as water. The controller is configured to operate the wiper in response to the output.

(8) Embodiments of the third aspect of the present disclosure can include any one or a combination of the following features: the window clearing system includes a pump for pressurizing cleaning fluid to apply to the windshield, wherein the controller is configured to selectively operate the pump in response to classification of the substance as debris.

(9) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. **1** is a perspective view of a vehicle incorporating a moisture detection system according to one aspect of the present disclosure;

(3) FIG. **2** is a functional block diagram of a moisture detection system for a vehicle according to one aspect of the present disclosure;

(4) FIG. **3** is an exemplary cross-sectional view of a vehicle incorporating an imaging device adjacent to a windshield of the vehicle within an interior of the vehicle for detecting moisture conditions in the region external to the vehicle;

(5) FIG. **4**A is an exemplary image captured by an imaging device in an interior of the vehicle positioned away from the windshield resulting in detectable water droplets on the windshield;

(6) FIG. **4**B is an exemplary image captured by an imaging device positioned within an interior of the vehicle and near the windshield demonstrating detection of water streaks on the windshield;

(7) FIG. **5** is a representation of a fully convolutional data description (FCDD) network processing an image captured by an imaging device positioned near the windshield and producing image data that demonstrates an obstruction in the image;

(8) FIG. **6**A-**6**C are captured images alongside image data that demonstrates obstructions on a windshield **18** of a vehicle after processing through a FCDD network;

(9) FIG. **7**A-**7**B are exemplary images captured by an imaging device for the vehicle demonstrating splash events along a roadway for the vehicle, with FIG. **7**B illustrating splash zones overlaying splash events;

(10) FIG. **8** is an exemplary process carried out by a moisture detection system according to one aspect of the present disclosure.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(11) Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements are not to scale and certain components are enlarged relative to the other components for purposes of emphasis and understanding.

(12) As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to a detailed design; some schematics may be exaggerated or minimized to show function overview. Therefore,

specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

(13) For purposes of description herein, the terms "upper," "lower," "right," "left," "rear," "front," "vertical," "horizontal," and derivatives thereof shall relate to the concepts as oriented in FIG. **1**. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(14) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to visibility obstruction detection for a vehicle **14**. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

(15) Referring generally to FIGS. **1-8**, a moisture detection system **10** uses image processing to detect conditions of a region external **12** to a vehicle **14**, such as water on an outer surface **16** of a windshield **18** of the vehicle **14** or water ahead of or around the vehicle **14**. In general, the present moisture detection system **10** may provide for enhanced space utilization within the vehicle **14** by allowing an imaging device **20** of the moisture detection system **10** to be positioned near or far from the windshield **18** while still detecting and classifying obstructions on the windshield **18**. Further, the moisture detection system **10** may provide for enhanced visibility by optimizing clearing of the windshield **18** and/or maneuvering, or proposing to maneuver, the vehicle **14** to a target location to enhance visibility and/or control. The moisture detection system **10** may further provide for more accurate detection of weather conditions, such as rain, humidity, fog, or other moisture conditions in the region external **12** to the vehicle **14**, thereby allowing for enhanced responsiveness for other vehicle systems.

(16) Referring now to FIG. **1**, the moisture detection system **10** for the vehicle **14** may incorporate an imaging device **20** positioned within a cabin **22** of the vehicle **14**. For example, the imaging device **20** may be positioned adjacent to the windshield **18** of the vehicle **14** within a passenger compartment **24** of the vehicle **14** and be oriented in a vehicle-forward orientation. At least one wiper **26** is positioned on the outer surface **16** of the windshield **18** for clearing the windshield **18** of water, debris, or other substances thereon. As will be described further with respect to FIG. **2**, a window cleaning system may be employed for cleaning the windshield **18** or any one of a plurality of windows of the vehicle **14** that may accumulate moisture or other substances on the window from the region external **12** to the vehicle **14**. The window cleaning system may include a nozzle **28** positioned adjacent to the wipers **26** between a hood **30** of the vehicle **14** and the windshield **18** of the vehicle **14** and be configured to spray cleaning fluid onto the windshield **18**. The cleaning fluid can include a cleaning agent, such as methanol, glycol, or other fluids for clearing substances off of the outer surface **16** of the windshield **18** when in use with the wipers **26**. It is contemplated that the cleaning fluid may incorporate water, which may be warmed to aid in a defrost operation of the windshield **18** in some examples.

(17) Still referring to FIG. **1**, a plurality of distance sensors **34**, **36**, **38** are incorporated in the vehicle **14** to detect other objects, such as other vehicles **14**, in the region external **12** to the vehicle **14**. The detection devices may include ultrasonic and/or infrared detectors, as well as cameras **38**, such as the imaging device, that are configured to detect distances from the vehicle **14** to surrounding objects in the region external **12** to the vehicle **14**. Accordingly, the images from the

imaging device **20** may be used to detect distance from the vehicle **14** to objects within a field of view of the imaging device.

(18) As will be described further with respect to FIG. **2**, the distance sensors **34**, **36**, **38** may include any one of the radio detection and ranging sensors (RADARs **34**), light detection and ranging sensors (LIDARs **36**), and cameras **38**. In some examples, ultra-wideband (UWB) sensors are employed for detecting objects in the region external **12**. In general, the distance sensors **34**, **36**, **38** may be configured to detect cross-traffic events and/or detect objects in blind spots of the vehicle **14** to aid a user of the vehicle **14** in maneuvering of the vehicle **14**. In the present example, the data collected by the detection sensors is used by the moisture detection system **10** to allow the moisture detection system **10** to detect distance (e.g., a following distance **120**) from the vehicle **14** to splash events **118** and to classify the importance, relevance, or priority, of the splash events **118** based on the distance.

(19) Still referring to FIG. **1**, the vehicle **14** includes a plurality of wheels **40** each having a tire **42** interacting with a driving surface **44** for the vehicle **14**. Friction between each tire **42** and the driving surface **44** may be influenced by moisture conditions between the tires **42** and the driving surface **44**. Accordingly, the moisture detection system **10** may be employed to control driving of the wheels **40** to enhance maneuverability of the vehicle **14** along the driving surface **44**.

(20) The vehicle **14** includes at least one lighting assembly **46**, such as a headlamp assembly having headlights **48** configured to illuminate the region external **12** to the vehicle **14**. For example, the lighting assemblies may be configured to illuminate the region external **12** to the vehicle **14** at a plurality of illumination levels (e.g., high beams, low beams, etc.). Control of the lighting assemblies and the power levels and/or the illumination levels thereof may be enhanced by the moisture detection system **10**. For example, moisture conditions in the region external **12** detected by the moisture detection system **10** may cause the moisture detection system **10** to control the power levels of the lighting assemblies due to reduced visibility from the moisture conditions.

(21) Referring now to FIG. **2**, the moisture detection system **10** includes an imaging system **50**, a distance detection system **52**, and a response control system **54** in communication with the imaging system **50** and the distance detection system **52**. In general, data from the imaging system **50** and/or the distance detection system **52** be communicated to the response control system **54**, and the response control system **54** may control one or more visibility features and/or motion control features of the vehicle **14**. For example, the response control system **54** includes one or more controllers **56**, **72** having at least one processor and memory in communication with the processor. The memory may store instructions that, when executed by the processor(s) of the response control system **54**, cause the response control system **54** to perform various tasks related to enhancing visibility of the region external **12** from the compartment of the vehicle **14** and/or motion control for the vehicle **14**. In general, one or more of the controllers **56**, **72** and/or other electrical components that provide for decision-making via software may be referred to as control circuitry.

(22) For example, a response controller **56** of the response control system **54** may include a motion control unit **58** and a visibility control unit **60**. The visibility control unit **60** may be configured to control the window clearing system, a light control system **62**, and/or any other system influencing visibility through one or more of the windows of the vehicle **14**. The motion control unit **58** may control communication to one or more vehicle systems, such as a powertrain **64** of the vehicle **14**, a brake system **66** of the vehicle **14**, or any other motion control system for the vehicle **14**. For example, the motion control unit **58** may include a speed sensor in the powertrain **64** that detects rotations of gearing in the powertrain **64**, however, other speed sensors may be used to detect or infer the speed of the vehicle (RF waves, inductive sensing, capacitive sensing, etc). Further, the response control system **54** may include and/or be in communication with a display **68**, such as a human-machine interface (HMI **70**) in the compartment of the vehicle **14**. The display **68** is configured to present messages to a user and/or allow the user to control the window clearing system, the light control system **62**, or any other aspect related to visibility and/or motion of the

vehicle **14**. In general, the response control system **54** may be configured to actively control visibility and/or motion features for the vehicle **14** or to passively present messages at the display **68** to indicate visibility and/or motion target operations for the user to initiate.

(23) With continued reference to FIG. **2**, the window clearing system can include a window clearing controller **72** that controls the operation of a pump **74** that pressurizes the cleaning fluid for spraying the cleaning fluid via the nozzle **28** onto the windshield **18** as previously described. A valve **76** may fluidly interpose the pump **74** and the nozzle **28** to selectively allow the cleaning fluid to enter the nozzle **28**.

(24) The window-clearing controller may further be in communication with a motor **78** that drives the wipers **26**. For example, the motor **78** may be configured to rotate the wipers **26** over the windshield **18** in response to signals from the window clearing controller **72**. In some examples, the speed of the vehicle **14**, as detected by the speed sensor, may be compared to detected splash events **122** and, in response to this detection, the motor **78** may be energized to operate at a particular number of rotations per minute (RPM). At least one switch **80** is in communication with the window clearing controller **72** and/or directly in communication with the motor **78**, the pump **74**, and/or the valve **76** to control dispensing of the cleaning fluid and/or driving of the wipers **26** via manual interaction. For example, the at least one switch **80** may include a first mechanism **82** that causes dispensing of the cleaning fluid on the windshield **18** and a second mechanism **84** that controls the operation of the wipers **26**. For example, the first mechanism **82** may be a knob that, when pulled or pushed, causes the cleaning fluid to be dispensed, and the second mechanism **84** may be a button or knob that causes the wipers **26** to move over the windshield **18**. It is contemplated that the window clearing controller **72** may be omitted in some examples and that the response control system **54** may directly control the window clearing operations. In such an example, at least one of the switches **80** interposes the valve **76** and the nozzle **28** (e.g., the first mechanism **82**), at least one of the switches **80** interposes the motor **78** and the wipers **26** (e.g., the second mechanism **84**). In either example, the user may manually control dispensing of the cleaning fluid and/or operation of the wipers **26**, and such operations may also or alternatively be automatically controlled by the response control system **54**. In another example, instructions to initiate wiping of the windshield **18** and/or cleaning of the windshield **18** are represented at the HMI **70**, and automatic control of the window clearing system is omitted.

(25) The visibility control unit **60** may also or alternatively be configured to control the lighting assemblies of the vehicle **14** based on moisture conditions detected by the imaging system **50**, as previously described. For example, in the event that it is raining in the region external **12** to the vehicle **14**, the headlights **48** may be automatically activated or the response control system **54** may communicate an instruction to present a message at the display **68** for the user to activate the headlights **48** in response to detection of moisture conditions in the region external **12** to the vehicle **14**. Further, brightness levels (e.g., binary high-beams/low-beams or fine brightness controls) may be controlled by the response control system **54** actively or passively (e.g., presentation of messages at the display **68**).

(26) In general, the motion control unit **58** may control the various systems of the vehicle **14** related to motion control, such as driving of the wheels **40** (e.g., torque values), brakes of the brake system **66** (e.g., traction control, antilock brakes (ABS)), steering of the vehicle **14** (e.g., maneuvering along the driving surface **44**), or any other motion control unit **58** for the vehicle **14**. Similar to the operation of the visibility control unit **60**, features of the vehicle **14** related to motion control may be presented at the display **68** in addition, or an alternative to, automatic control of maneuverability of the vehicle **14**. For example, in an at least partially semi-autonomous mode of control for the vehicle **14**, the visibility control unit **60** may communicate an indication to the user at the display **68** to reduce speed, maneuver the vehicle **14** to the left or the right, or the like, and, in an alternative, may control the speed, maneuverability to the left or the right, etc., in response to detection of the moisture conditions.

(27) With continued reference to FIG. **2**, the various responses communicated by the response control system **54** may be based on outputs from one or both of the imaging system **50** and the distance detection system **52**. With particular reference to the imaging system **50**, the imaging device **20** captures one or more images (captured images **86**) of the region external **12** to the vehicle **14**, which are then processed by an image processor **88** to detect moisture conditions on or around the vehicle **14** and produce one or more output images **90**. The image processor **88** is in communication with a memory **91** that stores instructions that, when executed, cause the image processor **88** to detect water streaks, water droplets **108**, splash events **118**, splash sources **116**, or any other optical distortion related to moisture condition detection and/or obstruction detection. Included with or in communication with the memory **91** is a fully convolutional data description network (FCDDNN **92**), which may be a neural network that segments the image data and detects optical distortions in the captured images **86**. The FCDDNN **92** may be trained by a training module **94** of the imaging system **50** that provides sample images and/or historical image data that presents optical distortion (e.g., optical distortion caused by moisture in an image). The FCDDNN **92** is employed to detect portions of the image that have distortion due to moisture conditions and will be described in further detail in reference to FIG. **5**.
(28) In addition to detecting moisture conditions, the image processor **88** is configured to detect any other obstruction within the field of view of the imaging device **20**. For example, the obstruction may be environmental debris, such as droppings, leaves, sticks, non-water material, or any other substance that may stick to or land on the outer surface **16** of the windshield **18**. The imaging system **50** may therefore distinguish between debris and water. For example, the FCDDNN **92** may be trained to score the obstruction with a level of opacity, light transmittance, light distortion, or the like. For example, debris may be associated with opacity, whereas water and/or other moisture conditions may be associated with bending of light through the obstruction.
(29) In general, distance information from the distance detection system **52** and moisture classification and detection from the image processing system may be used by the response control system **54** to initiate the vehicle **14** response. For example, if moisture is detected on the windshield **18** above a particular threshold (e.g., a low visibility threshold), the imaging system **50** may communicate an output to the response control system **54** indicating environmental conditions of the region exterior. For example, the imaging system **50** may determine that it is raining, snowing, or otherwise precipitating in the region external **12**, and communicate an indication to the response control system **54** to initiate the wipers **26**. The image processing system may further differentiate between various forms of precipitation and/or obstructions on the windshield **18** to allow for an individualized response initiated by the control system based on the environmental conditions. For example, and as will be described with respect to the foregoing figures, the imaging system **50** may classify types of obstructions as moisture-related or light-related, and the signal communicated from the imaging system **50** of the response control system **54** may be dependent on the type of obstruction.
(30) For example, the imaging system **50** may detect droppings on the windshield **18** and communicate an output for the response control system **54** to control the pump **74** of the window clearing system to spray the cleaning fluid onto the windshield **18**. In another example, the imaging system **50** detects spray on the windshield **18** from splashing from another vehicle **14** ahead of the vehicle **14** and, in response, communicates a signal to the response control system **54** to control, or communicates a message to the HMI **70** for the user to control, the vehicle **14** to slow or adjust the positioning of the vehicle **14**. The lighting assemblies may further be controlled in response to the detection of the moisture conditions to illuminate the region external **12** to the vehicle **14** at a particular power level/illumination level and/or to turn the headlights **48** on or off. These responses are exemplary and nonlimiting, such that any combination of responses may be performed by the moisture detection system **10** in response to classification by the imaging system **50**.
(31) Still referring to FIG. **2**, feedback from the window clearing system, the moisture control

system, the HMI **70**, or other vehicle systems may be monitored by the response control system **54** to optimize the response by the moisture detection system **10**. For example, upon a classification of the imaging system **50** of the environmental conditions in the region exterior, the imaging system **50** may communicate an instruction, or signal, to the response control system **54** that there is a light rain in the region external **12** to the vehicle **14**. In response to detection of the light rain, the response control system **54** may initiate wiper control at a low speed. However, manual control of the wipers **26** to stop wiping of the windshield **18** via the second switch **80** may end the automatic control of the wipers **26**. The response control may detect the feedback (e.g., the user manually shutting off the wipers **26**) and update, or optimize, future responses when detecting light rain. In this way, the detection and classification of the environmental conditions, including the moisture conditions, and the response thereto may be enhanced to promote optimized responses by the user. It is contemplated that other examples related to other environmental conditions may be similarly optimized using feedback (e.g., manual adjustment of the lighting assemblies, manual steering adjustments, braking adjustments, etc.). In some examples, dismissal of the messages presented at the HMI **70** by the user may indicate that the response indicated in the message is undesired. Other examples of feedback and control will be described with respect to the proceeding figures.

(32) Referring now to FIG. **3**, an exemplary arrangement of the imaging device **20** relative to the windshield **18** is illustrated to demonstrate the proximity of the imaging device **20** relative to the windshield **18** and the enhanced optical and spatial properties of the arrangement. Although illustrated as disposed adjacent to an upper portion **96** of the windshield **18**, it is contemplated that the imaging device **20** may disposed along any portion of any window of the vehicle **14** that allows the imaging device **20** to capture a view of the region external **12** to the vehicle **14**. In general, the software employed by the moisture detection system **10** of the present disclosure allows for the positioning of the imaging device **20** to be at least partially distance-independent between a lens **98** of the imaging device **20** and the window. For example, and as will be described with respect to FIGS. **4**A and **4**B, water on the windshield **18** (e.g., condensation, liquid water, frost) may appear different in the images depending on a distance of the imaging device **20** relative to the windshield **18**. For example, water droplets **108** may appear as discrete circles, dots, or other geometric shapes that are more clearly defined along edges **110** of the shapes when the imaging device **20** is spaced further from the windshield **18** relative to being against the windshield **18** or very close to (e.g., between 0 and 35 mm) the windshield **18**. The processes and methods carried out by the moisture detection system **10** of the present disclosure may allow for detection of the moisture conditions at either or both of a first position **102** of the imaging system **50** (e.g., at a first distance) and a second position **104** of the imaging device **20** (e.g., at a second distance less than the first distance). For example, the moisture on the windshield **18** may appear as blurry streaks or less defined optical distortions when an imaging device **20** is at the first distance to the windshield **18** due to the imaging device **20** being close to the water on the windshield **18** than when positioned at the second distance. By providing for a distance-independent arrangement, the moisture detection system **10** allows for enhanced spacing within the compartment, a larger and more detailed field of view for the imaging device, and a universal application to allow for imaging devices at different distances from the windshield **18**.

(33) With continued reference to FIG. **3**, the windshield **18** may extend at an oblique angle **106** relative to vertical and/or horizontal orientations. For example, relative to the driving surface **44**, the windshield **18** may slant upward. The oblique angle **106** of the windshield **18** may further distort or otherwise influence how the moisture conditions appear in the images captured by the imaging device. The distortion may be more pronounced in examples in which the imaging device **20** is in the second position **104** relative to the first position **102**. Accordingly, challenges of capturing moisture conditions of the environment exterior to the vehicle **14** from images captured from an imaging device **20** within the compartment may be greater than or different from image processing for moisture detection of images captured from imaging devices outside of the

compartment. For example, water droplets **108** on the windshield **18** may expand when landing on the windshield **18** to be more elongated than droplets **108** on a vertical, or more inclined, surface, such as the surface of a passenger window, a rear window, a cover for an exterior camera, etc.
(34) Referring now to FIGS. **4**A and **4**B, differences in captured images **86** from the first position **102** relative to the second position **104** of the imaging device **20** in FIG. **3** are demonstrated. As illustrated in FIG. **4**A (the first position **102**), rainwater on the outer surface **16** of the windshield **18** may appear as droplets **108** having discrete shapes or edges **110** more clearly defined than a blur **112** due to water conditions as demonstrated in FIG. **4**B (the second position **104**). Using the current algorithm and processes by the moisture detection system **10**, such optical distortion, though mild, may be detected. By way of example, a droplet detection algorithm may be suitable for detecting moisture conditions when the imaging device **20** is positioned far from the windshield **18**, and a blur **112** detection algorithm may be employed by the imaging system **50** to detect moisture conditions on the windshield **18** when the imaging device **20** is positioned closer to the windshield **18** (FIG. **4**B). Accordingly, the moisture detection system **10** may provide for enhanced spacing within the vehicle **14** by providing the imaging device **20** close to the windshield **18**.
(35) In some examples, the second position **104** is within 50 mm of the windshield **18**. In some examples, the second position **104** is between 0 mm and 35 mm from the windshield **18**. In some examples, the second position **104** is positioned between 0 mm and 10 mm from the windshield **18**. In each of these examples, the second position **104** is proximate to the windshield **18** to provide for an enhanced field of view and spacing within the compartment. In these examples, the first position **102** is further away from the windshield **18** than the second position **104**.
(36) Referring now to FIG. **5**, the FCDDNN **92** is demonstrated processing the captured image of FIG. **4**B to produce the corresponding output image. The FCDDNN **92** is a deep learning model designed for the purpose of unsupervised irregularity detection in data and operates by learning the underlying patterns and structures within the image data to distinguish normal patterns from anomalous patterns. The FCDD may include a plurality of layers **114**, with each layer being a convolutional layer. Because each layer may be convolutional, the FCDDNN **92** may capture spatial dependencies and maintain the spatial structure of the input data.
(37) In the FCDDN, the input data is passed through a series of convolutional layers **114**, which extract relevant features at different levels of abstraction. These convolutional layers **114** may be followed by pooling layers **114** to downsample feature maps and reduce spatial dimensions of the feature maps. The output of the convolutional layers **114** may then be flattened and fed into fully connected layers **114**, which perform further feature extraction and map the learned features to irregularity scores. Irregularity detection can be achieved by comparing the computed irregularity scores to a predefined threshold, where scores above the threshold indicate anomalous instances. In the present examples, the predefined thresholds may correspond to edge continuity of shapes in the image to detect distortion caused by moisture (e.g., blurriness). These thresholds may be actively adjusted based on the user feedback previously described (e.g., the user manually activating wipers **26** and/or the clearing process, the user manually activating the headlights **48**, etc.).
(38) Referring now to FIGS. **6**A-**6**C, exemplary captured images **86** from the imaging device **20** directed to the windshield **18** are depicted alongside filtered image data (output images **90**) that indicate moisture conditions on the windshield **18**. The imaging system **50** may process the captured images **86** in the image processor **88**, including processing the captured images **86** in one or more machine learning models and/or neural networks. The imaging system **50** may employ edge detection techniques, histogram equalization, linear filters, image segmentation, convolution, or any combination of any image processing techniques to detect at least one portion of the captured image having a distortion or an obstruction. For example, and with reference to FIGS. **6**A and **6**B, the imaging system **50** may detect the blur **112** on the windshield **18** corresponding to wetting conditions (e.g., moisture on the windshield **18**). In another example, the imaging system **50** detects non-moisture conditions, such as one of the wipers **26** moving across the windshield **18**

(FIG. **6**C). In these examples, detection may be applied to the captured images **86** to associate groups of pixels of the captured images **86** with surrounding pixels in order to classify the image data as moisture-related or not moisture related. For example, lane detection may be employed by the imaging system **50** to identify one or more lane lines defining the plurality of lanes **126**. In general, object classification may be performed by the imaging system **50**, such as classifying other vehicles **14**, streetlamps, trees, or any other object captured in the captured images **86**.

(39) In general, the FCDDNN **92** previously described with respect to FIG. **5** may be employed for providing a wetness estimate. Based on the wetness estimate, which may be a continuously changing value or a binary wet or non-wet value, the response control system **54** may determine the response of the moisture detection system **10**. By employing the various layers **114** of the FCDDNN **92** in combination with manual feedback from the user as previously described with respect to FIG. **2**, the techniques employed by the moisture detection system **10** may provide for enhanced reaction. For example, the wipers **26** may be activated when the user would ordinarily activate the wipers **26** or prior to when the user would ordinarily activate the wipers **26** based on the training of the FCDDNN **92**. Further, the speed of the wipers **26** and/or activation of the pump **74** for cleaning the windshield **18** may be optimized based on the moisture level and/or previous activations of the wipers **26** or the pump **74**. The headlights **48** of the vehicle **14** may also, or alternatively, be activated with limited false activations. For example, the FCDDNN **92** detects and classifies the image data as oncoming headlights **48** and differentiates oncoming headlights **48** from moisture conditions or non-moisture conditions. Further, objects having pronounced moiré effects (e.g., fences) may be differentiated by the FCDDNN **92**. In some examples, the manual feedback includes the manual operation of the wiper (e.g., via the at least one switch **80**) and/or maneuvering of the vehicle **14** away from a splash event **118**. Further, and as will be described below, the adjusting the vehicle **14** to another lane or maintaining positions in a current lane may be manual feedback (e.g., ignoring a recommendation to change lanes or manual override of moving into another lane).

(40) Referring now to FIGS. **7**A and **7**B, identification of at least one splash source **116** of splash events **118** may be performed by the imaging system **50** of the moisture detection system **10**. For example, the splash sources **116** may include vehicles **14** and non-vehicles **14**. The splash source **116** may be tires **42**, a vehicle body, bridges, overpasses, construction equipment, hydrants, or any other source. Accordingly, the imaging system **50** may be configured to classify the sources using any one of the previous image detection techniques previously described.

(41) With continued reference to FIGS. **7**A and **7**B, the imaging system **50** may further, or alternatively, determine a location of the splash source **116** relative to the vehicle **14**. Based on the location of the splash source **116** (e.g., a source lane **126***a*), the response control system **54** may control any one of the window cleaning system the lighting system, the HMI **70**, and/or the motion control system. For example, the imaging system **50** may locate the splash source **116** in another one of the plurality of lanes **126** other than the lane of the vehicle **14** and, based on location of the splash source **116** in another lane, may recommend maintaining course in the current lane of the vehicle **14**. Thus, in general, the moisture detection system **10** may control or indicate a message to control the vehicle **14** to adjust lanes based on the location of the splash source **116**. In the example, as illustrated in FIG. **7**B, multiple splash sources **116** are detected by the imaging system **50**, and the response control system **54** may recommend one of the plurality of lanes **126** based on proximity of the vehicle **14** to one or more of the plurality of splash events **118** (e.g., based on a following distance **120** of the vehicle **14** to the other vehicles **14**, based on relative position of other vehicles **14** to other sources of splash such as a pothole). Stated another way, the moisture detection system **10** may detect a first splash source **116** and a second splash source **116**, classify each splash source **116** with a relevance value, and recommend a lane or other location for the vehicle **14** based on the relevance values of the first and second splash sources **116**. As will be described below, the moisture detection system **10** may further recommend a target following distance, a target speed, or

the like in response to the relevance values.

(42) With particular reference to FIGS. **7**B, at least one splash zone **122** may be determined, or calculated, by the imaging system **50** based on image data from the initial images. For example, any one of the previous techniques described, including the application of the FCDDNN **92**, may allow the imaging system **50** to estimate dimensions of the splash zone **122**, such as a depth D, a width W, and a height H of the splash. The dimensions of the splash zones **122** may be determined based on the following distance **120** from the source of the splash to the vehicle **14** and the captured images **86**. For example, the following distance **120** may be determined using the distance sensors **34**, **36**, **38** previously described with respect to FIGS. **1** and **2**, and the forward-most location of the splash may be detected based on image processing of the captured images **86**. The difference between the following distance **120** and the front of the splash zone **122** may be calculated by the control circuitry to determine the depth D of the splash zone **122**. In this way, the distance detection and the location detection of the system **10** may provide for determining the origin point of the splash, the size of the splash, the likelihood of contacting the water of the splash based on distance and/or speed, and the like.

(43) In the present example, three splash zones **122** are detected by the imaging system **50**, with each of the splash zones **122** having a different size. The size of the splash zones **122** may be caused by the sizes of the tires **42**, the speeds of the vehicles **14**, the lanes that the vehicles **14** are in (e.g., road roughness, potholes, etc.) or any other factor that may influence the size of the tire splash. For example, uneven roadways that may cause puddles, curves in the road (e.g., a slope in the road, turns), or the like may further influence the size of the splash. Based on the size of the splashes, the imaging system **50** may determine the following distance **120** from each or any one of the splash sources **116** (e.g., the other vehicles **14**). In general, sizes of the splashes, a density of the splashes, a duration of the splashes, or any other aspect related to the magnitude of the splashes may be detected by the imaging system **50** and used to classify a priority or a ranking (e.g., the relevance values) of the splash events **118** to determine a target lane of the plurality of lanes **126**, a target following distance **120** (e.g., a minimum following distance) for the vehicle **14** relative to other vehicles **14**, activation of the window clearing system, presentation of messages at the HMI **70**, control of the vehicle **14**, or any of the responses previously described. The direction of the splashes may further be detected, which may be influenced by wind speed and direction, to enhance estimation of the splash zones **122** and further enhance the response determined by the response control system **54**. Such wind speed and direction may be detected from weather sensors or a Global Positioning System (GPS) in communication with the moisture detection system **10**.

(44) The moisture detection system **10** may further, or alternatively, estimate the density of moisture conditions, such as the density of the splash zone **122** or the density of rain. For example, based on the amount, distribution, or pattern of blurs **112** or other moisture spots detected on the windshield **18**, the moisture detection system **10** can estimate the density of rain and activate the wipers **26** or communicate an indication for the user to operate the wipers **26** in response to the amount, distribution, or pattern exceeding a threshold or matching a target distribution or pattern. For example, if more than 25%, 50%, or 90% of the windshield **18** in the field of view of the imaging device **20** has blurs **112**, the imaging system **50** may communicate a signal to the response control system **54** to initiate the wipers **26**. In other examples, the sizes of the water droplets **108** and/or blurs **112** may be categorized by the FCDDNN **92** and compared to stored moisture conditions to determine the response of the moisture detection system **10**.

(45) In some examples, the moisture detection system **10** may access the position of the vehicle **14** relative to the roadway type. For example, the GPS may provide the position of the vehicle **14**, thereby providing the roadway type (e.g., highway, city driving, etc.). In this way, the number of lanes, the motion direction, the construction and/or manner of operation of the roadway (e.g., one-way roads, interstate roads with median, highway with no median) may be determined. Accordingly, the moisture detection system **10** may prioritize some lanes over other lanes or adjust

following distances **120** based further on types of the roadways. For example, because the typical speed of a vehicle **14** on an interstate roadway than another roadway, the moisture detection system **10** may suggest a “slow lane” of two lanes based on moisture conditions detected on the interstate roadway.

(46) Although the source lane **126***a* of the plurality lanes **126** is illustrated as having the same traffic direction as the other of the plurality lanes **126** in the previously-described features, in some examples, the plurality of lanes **126** includes lanes having a first traffic direction and lanes having a second traffic direction. In this example, the imaging system **50** is configured to classify the splash source **116** as being in a lane having the first traffic direction and/or the splash source **116** being in a lane having the second traffic direction. For example, the imaging system **50** may detect oncoming headlights **48** in an adjacent lane and, in response to this detection, classify the adjacent lane as having an opposing traffic direction. Thus, splash from oncoming traffic may be compared to splashes of leading traffic, and the moisture detection system **10** may provide for enhanced response determination for a target lane for the vehicle **14** based on either or both splash events **118** in the first and second traffic directions. It is also contemplated that the distance detection system **52** previously described may be used to further determine the target lane for the vehicle **14** by tracking the following distance **120** from the vehicle **14** to the leading vehicle **14**.

(47) Referring now to FIG. **8**, an exemplary process **800** carried out by the moisture detection system **10** is demonstrated for use with splash detection and moisture condition detection on the windshield **18**. At step **802**, the imaging device **20** captures images of the region external **12** to the vehicle **14**. At step **804**, the image processor **88** detects an event in the region next interior to the vehicle **14**. For example, the event may be moisture conditions, such as splash events **118**, water being on the outer surface **16** of the windshield **18**, objects being on the windshield **18**, or any other event related to visibility obstruction or distortion. At step **806**, the event is classified as being associated with the windshield **18** (e.g., on the windshield **18**) or spaced from the windshield **18**. For example, if the event is a splash event **118** related to other vehicles **14** ahead of the vehicle **14**, as opposed to splashed water on the windshield **18** itself. It is contemplated that the splash event **118** may be classified both as an on-windshield **18** event and an off-windshield **18** event depending on the following distance **120** between the splash source **116** and the vehicle **14**. If the event is an obstruction on the outer surface **16** of the windshield **18**, the image processor **88** may classify the obstruction at step **808**. For example, the imaging system **50** may classify the obstruction as debris or water. Based on classification of the obstruction, at step **810**, the moisture detection system **10** determines a response. For example, the response may be to initiate the application of clearing fluid on the windshield **18** in the event that the obstruction is debris, such as moist debris, or activation or adjustment of the speed of one or more of the wipers **26** on the windshield **18** in the event of moisture conditions. At step **812**, an output is communicated to initiate the response determined at step **810**. It is contemplated that the response may alternatively be a more passive response, such as presenting a message at the display **68** to instruct a user to carry out one or more of the functions that may alternatively be automatically carried out.

(48) At step **814**, feedback in the form of manual adjustment or non-operation (e.g., the user not following a recommendation) is communicated to the response control system **54** and/or the imaging system **50** to further refine the response determined in future events. For example, if the user is instructed to activate the wipers **26**, and the user does not activate the wipers **26**, the target moisture levels for determining wiper activation may be increased to a threshold by the imaging system **50** to limit false responses for future calculations. Such threshold may be the threshold for the FCDDNN **92** previously described or another threshold.

(49) If the event is not related to conditions of the windshield **18**, as determined in step **806**, the process may continue to determine a location of a splash event **118** at step **816**. For example, using the captured images **86**, the imaging system **50** may detect the source and/or location (e.g., the splash lane **126***a*) of the splash events **118** caused around the vehicle **14** (e.g., tire splash). At step

818, the response is determined by the response control system **54**. For example, the response may be to adjust the following distance **120** between the splash source **116** and the vehicle **14** by reducing the speed of the vehicle **14**. In other examples, the response includes maneuvering of the vehicle **14** to another lane of a plurality of lanes **126**. In other examples, the response includes presenting instructions via messaging at the HMI **70** to indicate to the user to maneuver the vehicle **14**, to adjust speed of the vehicle **14**, or the like. Other examples of the response include adjustments to the window cleaning system, such as activation of the wiper **26**, adjustment of the speed of the wiper **26**, activation of the pump **74** for applying cleaning fluid, or the like. At step **820**, the output may be communicated to initiate the response. Similar to step **812**, at step **822**, feedback, in the form of action or inaction by the user to undo the response communicated by the response control system **54**, is returned to the moisture detection system **10** for enhancing response determination in future conditions in which splash events **118** are detected.
(50) In general, the present moisture detection system **10** enhances responses for the vehicle **14** to limit obstruction and/or distortion of visibility of the region external **12** to the vehicle **14**. The image processing techniques employed by the moisture detection system **10** may enhance the room in the interior of the vehicle **14** by allowing the imaging device **20** to be positioned proximate to the windshield **18**. Further, the image processing techniques employed herein may provide for enhanced detection of splash sources **116** and/or moisture conditions on the outer surface **16** of the windshield **18**. Based on the detection of these moisture events, quick response times for clearing of the windshield **18** and/or optimize maneuvering of the vehicle **14** may be provided by the moisture detection system **10**.
(51) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.
(52) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.
(53) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an endpoint of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or endpoint of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.
(54) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other,

such as within about 5% of each other, or within about 2% of each other.
(55) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.
(56) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further, it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle, comprising: a window separating a compartment of the vehicle from a region external to the vehicle; an imaging device in the compartment and configured to capture an image of the region external to the vehicle through the window; control circuitry in communication with the imaging device, the control circuitry configured to: receive the image from the imaging device; process the image in a fully-connected data description neural network (FCDDNN) trained to determine a moisture level; detect an obstruction in the image; correlate the obstruction with a substance on the window; classify the substance as water or debris based on the moisture level; and generate an output in response to detection of the obstruction.
2. The vehicle of claim 1, further comprising: a wiper selectively moveable along the window; and a window clearing system including a controller in communication with the control circuitry and configured to operate the wiper in response to the output.
3. The vehicle of claim 2, wherein the window clearing system includes a pump for pressurizing cleaning fluid to apply to the window, wherein the controller is configured to selectively operate the pump in response to the output.
4. The vehicle of claim 3, wherein the output generated by the control circuitry is based on the classification of the obstruction as water or debris.
5. The vehicle of claim 4, wherein the controller is configured to operate the wiper in response to classification of the substance as water and operate the pump in response to classification of the substance as debris.
6. The vehicle of claim 4, wherein the imaging device is disposed within 50 millimeters (mm) from the window.
7. The vehicle of claim 3, wherein the control circuitry includes a training module that trains the FCDDNN based on manual operation feedback.
8. The vehicle of claim 7, wherein the manual operation feedback includes manual operation of at least one switch controlling at least one of the wiper and the pump.
9. The vehicle of claim 7, wherein the training module is configured to train the FCDDNN to differentiate between light sources in the region external and the substance on the window.
10. The vehicle of claim 1, wherein the control circuitry is configured to classify a level of optical distortion with the moisture level of the substance.
11. The vehicle of claim 1, wherein the window is a windshield of the vehicle extending at an oblique angle relative to a lens of the imaging device.
12. The vehicle of claim 1, wherein the imaging device is disposed within 15 millimeters of the window.
13. The vehicle of claim 1, wherein the control circuitry is configured to: detect another vehicle ahead of the vehicle; and classify the substance as tire splash from the other vehicle.
14. The vehicle of claim 1, further comprising: an automatic speed control system for the vehicle in communication with the control circuitry, wherein the automatic speed control system is configured to control the speed of the vehicle based on classification of the substance as tire splash from

another vehicle.
15. A vehicle, comprising: a windshield separating a compartment of the vehicle from a region external to the vehicle; an imaging device in the compartment and configured to capture an image of the region external to the vehicle through the windshield; and control circuitry in communication with the imaging device and configured to: receive the image from the imaging device; detect an obstruction in the image corresponding to a substance on the windshield; detect another vehicle ahead of the vehicle; classify the substance as tire splash from the other vehicle; and generate an output in response to the detection of the obstruction, wherein the imaging device is disposed within 50 millimeters (mm) from the windshield.
16. The vehicle of claim 15, further comprising: a wiper selectively moveable along the windshield; and a window clearing system including a controller in communication with the control circuitry and configured to operate the wiper in response to the output.
17. The vehicle of claim 15, wherein the imaging device is disposed within 15 mm of the windshield.
18. The vehicle of claim 15, wherein the control circuitry is configured to classify a level of optical distortion with the moisture level of the substance.
19. A vehicle, comprising: a windshield separating a compartment of the vehicle from a region external to the vehicle; an imaging device in the compartment and configured to capture an image of the region external to the vehicle through the windshield; and control circuitry in communication with the imaging device, the control circuitry configured to: receive the image from the imaging device; process the image in a fully-connected data description neural network (FCDDNN) trained to determine a moisture level; detect an obstruction in the image; correlate the obstruction with a substance on the windshield; detect another vehicle ahead of the vehicle; and classify the substance as tire splash from the other vehicle.
20. The vehicle of claim 19, wherein the control circuitry is configured to control a wiper based on the classification of the substance as water.